Interrupts and Interrupt Vector Table in 8086

Lecture Notes

1 Interrupt

An **interrupt** is a signal that temporarily stops the execution of the current program so that the processor can execute a special routine called the **Interrupt Service Routine (ISR)**.

Execution Flow

```
Program executing

Interrupt occurs

CPU executes ISR

CPU returns to program
```

Important point: The processor always **completes the current instruction first** and then responds to the interrupt.

2 Interrupt Types in 8086

The 8086 processor supports **256 interrupt types**.

$$2^8 = 256$$

Range of interrupt types:

$$00H \text{ to } FFH$$

Interrupts may be classified as:

- Hardware interrupts
- Software interrupts

3 Interrupt Vector Table (IVT)

The **Interrupt Vector Table** stores the starting addresses of all Interrupt Service Routines.

Location of IVT

$$00000H \text{ to } 003FFH$$

Total size

$$256 \times 4 = 1024 \text{ bytes} = 1 \text{ KB}$$

Each interrupt type uses **4 bytes** in the IVT.

4 Structure of an IVT Entry

Each interrupt vector contains four bytes arranged as follows:

Byte	Content
1	IP (low byte)
2	IP (high byte)
3	CS (low byte)
4	CS (high byte)

Where:

- **IP** = Instruction Pointer
- **CS** = Code Segment

These two registers together give the starting address of the ISR.

5 CS and IP Registers

Register	Meaning
CS	Base address of code segment
IP	Offset of instruction inside the segment

Both registers are **16-bit registers**.

6 Physical Address Formation

The physical address in 8086 is calculated using:

$$\text{Physical Address} = CS \times 10H + IP$$

Example

$$CS = 4A3C$$

$$IP = 332A$$

$$4A3C0 + 332A = 4D6EAH$$

Thus the instruction is stored at memory location:

$$4D6EAH$$

7 Finding the Location of an Interrupt Vector

To find the location of an interrupt vector in the IVT:

$$\text{IVT Address} = \text{Interrupt Type} \times 4$$

8 Numerical Example

Problem

Find the ISR address for interrupt type:

$$42H$$

Memory contents are given as:

Address	Value
108	2A
109	33
10A	3C
10B	4A

8.1 Step 1: Find IVT Location

$$\text{IVT Address} = 42H \times 4$$

$$= 108H$$

Therefore the ISR address starts at memory location:

$$000108H$$

8.2 Step 2: Read Four Bytes

Address	Value
108	2A
109	33
10A	3C
10B	4A

8.3 Step 3: Form IP

First two bytes represent IP (little endian format).

$$\text{Low byte} = 2A$$

$$\text{High byte} = 33$$

Therefore

$$IP = 332A$$

8.4 Step 4: Form CS

Next two bytes represent CS.

$$\text{Low byte} = 3C$$

$$\text{High byte} = 4A$$

Therefore

$$CS = 4A3C$$

8.5 Step 5: Calculate Physical Address

$$\text{Physical Address} = CS \times 10H + IP$$

$$= 4A3C0 + 332A$$

$$= 4D6EAH$$

9 Final Result

$$CS = 4A3C$$

$$IP = 332A$$

$$\text{ISR Physical Address} = 4D6EAH$$

10 Interrupt Execution Sequence

1. Interrupt occurs
2. CPU completes current instruction
3. CPU pushes FLAGS, CS, and IP onto the stack
4. CPU loads new CS and IP from IVT
5. Interrupt Service Routine executes
6. IRET instruction returns control to the original program

11 Summary

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Interrupt  IVT  CS:IP  ISR  IRET  Program continues
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